Abstract
This project proposes a unified framework for advancing “Space Debris Consciousness” - a cultural, technological, and policy-driven approach to orbital sustainability. The first component develops public urgency and awareness through gamified educational tools that visualize the growth and risk of orbital debris, drawing inspiration from global environmental movements such as the Earthshot Prize and the Astra Carta. The second component introduces an AI-based decision-making model adapted from domestic robotics, mapping the logic of household tidying to orbital debris prioritization. By modeling debris interactions as a Partially Observable Markov Decision Process (POMDP), this framework lays the foundation for intelligent debris avoidance and long-term autonomous mitigation. The project bridges human perception, policy formation, and machine autonomy, offering a scalable path from public awareness to AI-driven action. This project represents an early step toward my long-term research vision: developing robotic systems capable of tidying both Earthly and orbital environments.

1. Introduction & Motivation

The growing accumulation of orbital debris has become one of the most urgent and least publicly understood challenges in contemporary aerospace. Thousands of inactive satellites, fragmented spacecraft, rocket bodies, and collision remnants now occupy Earth’s orbital environment, creating a hazardous ecosystem that threatens current and future space operations, and the satellite infrastructure modern society relies upon. While government agencies, commercial operators, and research institutions increasingly acknowledge the severity of the problem, public awareness remains limited. Unlike climate change - where decades of advocacy, education, and cultural storytelling have generated widespread environmental consciousness - space debris lacks a comparable sense of urgency in the public imagination. Yet its long-term consequences are equally profound, beginning with life-threatening risks such as weakened disaster-response systems, reduced ability to track hurricanes, wildfires, and floods, and degraded climate-monitoring that protects communities. These impacts cascade into daily life: disrupting GPS navigation, global communication networks, weather forecasting, satellite internet and TV, agricultural monitoring, and even the timing systems that support financial transactions.

This project begins with the central question: How do we build a global culture of “space debris consciousness,” in which the public, policymakers, and industry recognize orbital sustainability as a shared responsibility? Inspired by the trajectory of environmental movements and initiatives such as the Astra Carta and the Earthshot Prize, this work proposes a framework that connects awareness, policy, and technology. If people cannot see danger, they cannot care about it; If they cannot care about it, they cannot support the policies or technologies needed to solve it. Thus, the first phase of this work explores the role of gamified tools and interactive simulations in helping people *visualize* orbital debris and *experience* its risks in an intuitive, accessible way.

However, awareness alone is insufficient. My work in robotic decision-making (AA228) for autonomous housekeeping robots introduced me to the power of POMDPs and AI-driven controllers for operating under uncertainty. The logic of tidying a home - identifying items, assessing priorities, and acting efficiently in a cluttered, stochastic environment, maps naturally onto orbital debris management. Satellites in orbit must make decisions about which hazards to avoid, which paths to take, and how to operate safely with limited information and limited resources.

This convergence motivates the second phase of this project: the adaptation of decision-making frameworks from domestic robotics to orbital debris avoidance. By treating debris interactions as a partially observable control problem, we can begin exploring how AI can support improved collision avoidance, risk assessment, and long-term sustainability strategies. This linkage between human awareness and machine cognition forms the core of the "Space Debris Consciousness" concept: a continuum from perception → understanding → intelligent action.

Ultimately, this work also aligns with my long-term research goal to contribute to the field of specialty aerospace and robotics by developing robotic systems capable of sustaining environments, whether in a home or in orbit. While full-scale debris removal is a future goal, the first steps begin here, with conceptual foundations that bridge public engagement, AI-powered decision-making, and sustainable orbital policy. This project represents the earliest seed of that vision: a roadmap for how awareness and autonomy can work together to support a cleaner and safer space environment.

2. Background: The Space Debris Problem

Earth's orbital environment is rapidly becoming a congested and fragile ecosystem. Over 36,000 trackable pieces of debris larger than 10 cm and countless smaller fragments now populate low Earth orbit, medium Earth orbit, and geosynchronous orbit. These objects include inactive satellites, upper stages, collision fragments, and mission-related materials left behind across decades of space activity. Traveling at velocities of 7–8 km/s, even a 1 cm fragment can disable or destroy an operational spacecraft.

This increasingly complex environment raises the risk of multi-object collision chains known as the Kessler Syndrome, in which a single breakup event generates fragments that trigger additional collisions. Recent events, including the 2007 Chinese ASAT test and the 2009 Iridium–Cosmos collision, demonstrate how quickly debris populations can escalate. With the rise of commercial mega-constellations, the frequency of close approaches ("conjunctions") has increased to tens of thousands per week, highlighting the inadequacy of current tracking and coordination practices.

On the policy side, multiple organizations have attempted to establish frameworks for sustainable space operations. NASA, ESA, and the Inter-Agency Space Debris Coordination Committee have released mitigation guidelines, while global bodies such as UNCOPUOS have introduced voluntary sustainability standards. Meanwhile, initiatives like the UK's Astra Carta, led by King Charles III, attempt to elevate space sustainability into the cultural and ethical domain, akin to how environmentalism matured into a mainstream movement.

However, a significant gap remains between technical knowledge, policy frameworks, and public understanding. Unlike climate change, where public consciousness drives policymaking and funding, space debris remains invisible to non-specialists. Without accessible educational tools or emotional narratives, the problem feels distant, abstract, and disconnected from everyday life.

This disconnect highlights a crucial need for an integrated model: one that combines technical accuracy with public engagement, linking awareness to improved policy and enabling new technological solutions such as AI-assisted debris avoidance and eventual robotic cleanup. Understanding this background sets the stage for a framework that spans human perception, machine cognition, and long-term orbital stewardship.

3. Phase 1, Part 1- Public Awareness & Gamified Urgency

Despite its growing severity, the orbital debris problem remains largely invisible to the general public. Unlike environmental degradation on Earth, where images of melting ice caps, polluted oceans, or severe weather events create intuitive urgency, space debris is abstract, distant, and difficult to visualize. Yet the decisions humanity makes over the next two decades will determine whether low Earth orbit remains accessible or evolves into a high-risk environment dominated by cascading collisions and escalating operational costs. Bridging this awareness gap requires tools that translate orbital dynamics into human experience. This section proposes a gamified learning platform as the first phase in developing "space debris consciousness."

Gamification offers a powerful mechanism for demystifying complex scientific concepts. By transforming orbital interactions into interactive visualizations, simulations, and decision-based scenarios, users can directly experience how debris forms, spreads, and threatens satellites. The educational literature shows that engagement and comprehension increase dramatically when learners interact with dynamic systems rather than passively reading about them. In the context of space sustainability, gamified platforms can communicate risk in ways that charts or policy documents cannot: users can see debris multiply after a breakup event, observe how conjunctions evolve over time, and understand why even small fragments pose significant hazards.

A proposed platform would include several modules. First, a Debris Cascade Simulator, where users can trigger hypothetical events (such as a satellite explosion or collision) and observe how fragments propagate over days, weeks, and months. Second, a Conjunction Decision Game, in which players act as satellite operators making avoidance decisions based on incomplete or noisy data-mirroring real-world challenges. Third, an Orbit Stewardship Mode, inspired by conservation and environmental games, where users manage a “clean orbit,” track compliance with deorbiting rules, and receive rewards for responsible actions. Finally, a global leaderboard could highlight “top orbital stewards,” encouraging repeated engagement and social sharing.

To make this concrete, imagine a user experience:

- The screen opens on a visually stunning, real-time simulation of Low Earth Orbit, populated with thousands of dots representing active satellites and debris.
- A notification flashes: "Conjunction Alert! Satellite A and Debris Object B-2234. Probability of Collision: 1.2%." The user, acting as a satellite operator, has limited fuel and must decide: maneuver or wait?
- They choose to wait. A week later, a catastrophic collision occurs between two other satellites they weren't monitoring - an event they *triggered* indirectly by ignoring a minor debris event months ago in the simulator. The screen shakes as a cloud of new debris fragments radiates out, threatening their own satellite and others.
- This moment is important. The abstract risk of the Kessler Syndrome becomes a tangible consequence of their decisions. They immediately click "Restart," armed with a new understanding of orbital fragility and the interconnectedness of all objects in space.
- Complementary games would also be developed, focusing on different angles, such as the long-term impact of debris on daily human life (e.g., losing GPS, weather data) and strategic missions to actively reduce debris populations.

This platform would be web-based and mobile-first, ensuring maximum accessibility. It could use simplified orbital mechanics (like the game "Kerbal Space Program") for intuitive playability, while grounding its core dynamics in real physics and data from agencies like ESA and NASA.

Beyond education, gamification can serve as a cultural catalyst. Large-scale environmental movements such as climate activism grew when complex scientific predictions were transformed into stories, visual metaphors, and public challenges. The Earthshot Prize demonstrates how framing a planetary problem as an inspiring call to action can generate global momentum. Similarly, gamified space debris platforms, extended into deeper immersive VR/AR experiences, could anchor a broader awareness campaign, including school outreach kits, museum exhibits, public challenges ("Clear the Orbit Week"), and partnerships with organizations aligned with space sustainability, such as the Astra Carta.

This awareness layer is critical because policy shifts rarely occur without public understanding. In the early days of climate advocacy, policymakers faced resistance until public concern reached sufficient scale to justify regulatory intervention. Space debris is now at a similar inflection point: while experts warn of rising risks, widespread comprehension remains limited. A gamified platform could help build the public consciousness necessary to support new policies, fund mitigation technologies, and encourage responsible industry behavior.

Phase 1 therefore serves as the foundation for the entire Space Debris Consciousness framework. It transforms an invisible technical problem into an accessible human narrative, preparing the cultural and political environment for advanced decision-making tools (Phase 2) and autonomous mitigation systems (Phase 3). By enabling people to *see* and *feel* the orbital environment, we create the conditions for meaningful, sustainable action.

4. Phase 1, Part 2: Policy Framework for Orbital Sustainability

Addressing orbital debris requires not only technological innovation but a robust policy ecosystem that guides global behavior in space. Unlike terrestrial environments, Earth's orbital domain is a shared commons-used by all nations, owned by none, and regulated primarily through voluntary guidelines and national licensing requirements. This unique context has resulted in a governance landscape marked by fragmented authority, limited enforcement, and growing tension between commercial expansion and environmental sustainability. A comprehensive policy framework is therefore essential to support the cultural, AI-based, and robotic strategies proposed throughout this project.

4.1 Global Governance: UNCOPUOS and International Guidelines

The United Nations Committee on the Peaceful Uses of Outer Space (UNCOPUOS) remains the central forum for space governance. Its *Long-Term Sustainability (LTS) Guidelines* provide recommendations on debris mitigation, space traffic coordination, data sharing, and environmental responsibility. However, these guidelines are non-binding, leaving enforcement to individual states. The slow pace of multilateral agreement and the absence of penalties for non-compliance limit the effectiveness of global governance, especially as satellite constellations scale rapidly.

4.2 Technical Foundations: IADC Standards and ESA Clean Space

The Inter-Agency Space Debris Coordination Committee (IADC) serves as the technical backbone for debris mitigation policy. Its guidelines influence national regulations and industry best practices, including recommendations for post-mission disposal, probability-of-collision thresholds, and shielding requirements. ESA's Clean Space initiative extends this by promoting eco-design, end-of-life planning, and active debris removal concepts. These technical standards form the scientific basis for responsible operations.

4.3 National Regulations and Enforcement Gaps

Individual nations regulate their domestic operators through licensing.
Examples include:

- United States (FCC & NOAA): disposal requirements, casualty risk thresholds, collision avoidance reporting.
- United Kingdom: emerging sustainability requirements aligned with the Astra Carta.
- Japan & Europe: increasing focus on end-of-life passivation and trackability.

Yet the lack of globally harmonized standards leads to inconsistent practices, and many nations cannot enforce high standards on foreign operators whose debris affects everyone. The result is a governance patchwork unable to keep pace with escalating launch rates.

4.4 Ethics, Culture, and the Astra Carta

Recognizing the cultural and ethical dimensions of orbital sustainability, the United Kingdom introduced the Astra Carta in 2022. Championed by King Charles III, it frames space as an environment requiring stewardship, responsibility, and long-term thinking-similar to early environmental movements on Earth. The Astra Carta is not a regulation but a cultural scaffold, intended to elevate space sustainability into the global consciousness. It provides a moral foundation that aligns with the goals of "space debris consciousness."

4.5 Policy Gaps: Where Awareness and AI Become Crucial

Despite numerous guidelines, major gaps remain:

- No binding global regulations
- No unified space traffic management system
- Limited public pressure for accountability
- Unclear liability for active debris removal
- Absence of mechanisms to integrate AI-based decision systems
- Insufficient funding for cleanup missions

These gaps reveal a critical insight: technical solutions cannot succeed without public understanding and political will. Phase 1 (awareness) enables this cultural shift. Phase 2 (AI decision-making) provides the operational intelligence missing from current frameworks. Phase 3 (autonomous robotics) addresses long-term sustainability.

4.6 Public–Private Participation Through an Earthshot-Style Mobilization Framework

While awareness and policy lay the foundation for orbital sustainability, meaningful progress also requires a mechanism for broad participation and creative problem-solving. The Earthshot Prize offers a compelling model: it identifies major environmental challenges, invites global submissions, showcases innovative solutions, and elevates public consciousness through recognition and storytelling.

A **similar framework for orbital sustainability** could play a crucial role in translating cultural awareness into tangible action. Such an initiative would not focus on commercial ventures, but rather on **solutions**. A prize model could:

- identify key challenge areas (e.g., collision avoidance, debris characterization, trackability, end-of-life disposal, active debris removal)
- invite contributions from researchers, engineers, students, nonprofits, and international teams
- spotlight promising solutions through awards and global communication campaigns
- motivate interdisciplinary collaboration
- strengthen public understanding of orbital stewardship
- accelerate the development and testing of innovative approaches

This mobilization layer acts as a bridge between Phase 1 (awareness and policy literacy) and the more technical phases involving AI-assisted navigation and autonomous debris removal. It provides a structured pathway for public engagement, meaningful participation, and global cooperation.

In future work, I hope to explore how such a model could be adapted for space sustainability and tailor it to the unique challenges of the orbital environment.

4.7 Role of This Framework in the Policy Landscape

This project proposes an integrated model that supports policymakers, industry actors, and researchers by connecting three domains:

- Awareness and policy literacy establish the cultural and ethical grounding for responsible space activity while public–private mobilization transforms this awareness into participation and solution-generation across governments, research institutions, NGOs, and industry.
- AI supports compliance, improves traffic management, and reduces collision risk
- Robotics provides the physical capability for debris removal

In this way, the policy section completes the argument: orbital sustainability cannot be solved through isolated approaches. It requires a coordinated blend of cultural, regulatory, AI-driven, and robotic strategies

5. Phase 2 - AI Decision-Making for Debris Avoidance

While public awareness establishes the cultural foundation for space sustainability, technical progress depends on the ability to make informed decisions in an uncertain and dynamic orbital environment. Collisions do not simply occur because debris exists; they occur because satellites, rockets, and active spacecraft must continuously choose how to move through an imperfectly understood field of high-speed objects. This decision-making problem closely resembles the challenges addressed in autonomous robotics-particularly in cluttered environments where agents must identify hazards, prioritize tasks, and act efficiently with limited information. In this phase, we adapt the decision-making logic used in domestic tidying robotics to the orbital debris domain, proposing an AI-based framework to support avoidance, prioritization, and, eventually, mitigation.

At the core of this approach is the recognition that orbital debris management is a partially observable problem.Satellite operators do not perfectly know the location or trajectory of all debris.

Tracking catalogs contain uncertainties; smaller fragments are often untracked; conjunction predictions shift as new data becomes available. This mirrors the structure of a Partially Observable Markov Decision Process (POMDP) - the same framework used for autonomous ground robots that must reason under uncertainty about cluttered environments.

In a household tidying robot, as explored in AA228, the agent must decide which objects to pick up, which direction to move, and how to optimize its path based on noisy perception and limited time. Similarly, an autonomous spacecraft or debris-aware flight controller must determine which objects pose real collision risks, how to sequence avoidance maneuvers, and how to minimize fuel expenditure while maintaining mission objectives. By reframing orbital debris interactions as a POMDP, we establish a computational structure for making intelligent, optimized decisions.

A simplified model would define:

- States: Satellite position, debris positions (with uncertainty), relative velocities.
- Actions: Maneuver, hold position, adjust attitude, gather more tracking data.
- Observations: Sensor measurements, ground-based tracking updates, uncertainty distributions.
- Rewards:
    - Large positive rewards for maintaining safe distance,
    - Large negative rewards for collisions or near misses,
    - Small penalties for fuel use,
    - Additional penalties for unnecessary maneuvers.

A critical function of this POMDP framework is to reason explicitly about this uncertainty. The AI agent must not only decide on maneuvers but also determine when the uncertainty in a debris object's track is so high that the optimal action is to *gather more data*, for instance, by tasking a ground-based telescope or waiting for the next tracking update, rather than perform a premature, fuel-costly maneuver. This makes the system robust to the current limitations of the Space Situational Awareness (SSA) network, effectively treating data imperfection as a core part of the problem to be solved, not an external flaw.

This framework enables an AI agent to balance safety with mission endurance. It also provides a method for understanding when a maneuver is justified and when it is more efficient to wait for updated tracking. As satellite constellations grow in scale, manual decision-making becomes untenable, operators already face thousands of conjunction alerts per week. AI-based policies can automate much of the low-level reasoning, enabling human operators to focus on high-level oversight.

Integrating this decision-making layer with Phase 1 also strengthens public and policy understanding. Gamified simulations can allow users to "act" as satellite operators, making avoidance choices under uncertainty. This demonstrates not only the complexity of debris management, but also the importance of developing intelligent systems that can handle such complexity reliably.

The long-term vision extends beyond avoidance. The same decision-making models can be adapted for active debris removal missions: determining which objects to capture first, how to plan fuel-efficient trajectories, and how to sequence multi-object cleanup campaigns. In this sense, decision-making becomes the cognitive backbone for both prevention and removal-the logic that enables robotic systems in Phase 3 to act autonomously and intelligently.

By introducing AI decision-making into the framework of space debris consciousness, we bridge the gap between awareness and action. This phase shows how abstract understanding transforms into computational strategies that improve orbital safety today and prepare the foundation for autonomous cleanup systems tomorrow. In doing so, Phase 2 establishes the technological core of an integrated, sustainable approach to orbital debris management.

6. Phase 3 - Toward Autonomous Robotic Mitigation (Future Work)

While awareness (Phase 1) and AI-driven decision-making (Phase 2) lay the cultural and computational foundation for sustainable orbital environments, the long-term objective of space debris stewardship requires systems capable of acting autonomously to reduce existing debris populations. Phase 3 represents the culmination of this continuum: the development and deployment of robotic platforms that can identify, approach, capture, and deorbit hazardous objects.

Robotic debris removal presents distinct challenges: high relative velocities, diverse object shapes, uncontrolled tumbling, fuel constraints, and the legal complexities surrounding ownership of defunct satellites. Yet despite these difficulties, recent advancements in autonomous navigation, robotic manipulation, adaptive control, and on-orbit servicing demonstrate that the technology is rapidly approaching feasibility. Several missions-such as ESA's RemoveDEBRIS, Astroscale's ELSA-d, and JAXA's work with magnetic capture systems-indicate proof-of-concept pathways for active debris removal. However, these missions remain highly constrained demonstrations rather than scalable, repeatable solutions.

The integration of decision-making frameworks from Phase 2 provides a pathway toward scalable autonomy. A robotic cleanup system must prioritize which debris objects to remove, plan fuel-optimal rendezvous trajectories, and adapt to uncertain or changing dynamics of target objects. These requirements align naturally with POMDP-based reasoning and reinforcement learning policies, enabling robots to evaluate trade-offs, manage risk, and sequence actions for maximum mitigation impact. The same control logic that guides a domestic robot in navigating clutter can guide an orbital robot through chaotic debris fields-albeit at vastly different scales and physical regimes.

A conceptual architecture for such a system would include:

- Perception and Tracking Module: Fusing onboard sensors with ground-derived catalogs to maintain probabilistic estimates of debris states.
- Decision Layer (Phase 2): Computing action policies that determine when and how to approach, avoid, or capture debris objects.
- Robotic Manipulation Module: Robotic arms, nets, tethers, magnetic systems, or ion-shepherding propulsion for non-contact momentum transfer.
- Autonomous Mission Planner: Sequencing multi-object campaigns to maximize debris mass removed per unit of fuel and time.
- Human Oversight Loop: Operators setting high-level goals while AI executes low-level operations safely and repeatably.

This layered architecture directly mirrors that of intelligent domestic robotics. Instead of toys or clothing, the system interacts with derelict satellites. Instead of cluttered rooms, it navigates cluttered orbital shells. The analogy reinforces a key insight of this project: the logic of tidying is universal. Whether on the ground or in orbit, autonomous systems must sense their environment, infer uncertainties, decide wisely, and act efficiently.

In the long term, fully autonomous debris removal systems could operate continuously, becoming a kind of "orbital sanitation service" that maintains the safety and usability of Earth's orbital lanes. The cultural momentum generated in Phase 1 supports public and governmental willingness to fund such missions, while the technological backbone from Phase 2 ensures their feasibility and safety. Together, these phases form a complete pathway from understanding → policy → intelligent avoidance → active removal.

Phase 3 therefore represents the aspirational horizon of the Space Debris Consciousness framework. It offers a vision consistent with the future trajectory of this research: robots that safeguard our living environments on Earth and ensure the long-term sustainability of the orbit above us. This future work section outlines a roadmap for research, development, and implementation, serving as a bridge between present educational initiatives and the next generation of sustainable space technologies.

6. Impact, Challenges, and Contributions

The integrated "Space Debris Consciousness" framework aims to shift the trajectory of orbital sustainability by addressing the problem across cultural, technical, and operational dimensions. It establishes a scalable pathway from public understanding to intelligent policy to autonomous action. This section synthesizes the project's potential impact, acknowledges its challenges, and summarizes its core contributions.

6.1 Potential Impact

Cultural and Educational Impact
Phase 1 transforms orbital debris from an abstract scientific problem into an intuitive, emotionally resonant experience. Through gamified engagement and policy education, it fosters a new cultural norm - "space debris consciousness" - akin to the early public awareness that propelled the environmental movement. This shift is foundational, creating an informed citizenry and the political will necessary to support ambitious sustainability policies and fund advanced mitigation technologies.

Technical and Operational Impact
Phase 2 introduces an AI-driven decision-making architecture that directly enhances the safety and efficiency of space operations. By applying POMDPs and reinforcement learning, it provides a systematic framework for handling the uncertainty inherent in orbital debris fields. This model improves traditional workflows by:

- Reducing false alarms and unnecessary maneuvers,
- Optimizing fuel efficiency (ΔV expenditure),
- Learning adaptive strategies for conjunction events, and
- Providing operators with interpretable, probabilistic predictions.

As satellite constellations proliferate, this approach is critical for managing the escalating cognitive load on human operators, establishing a scalable foundation for semi-autonomous and autonomous collision avoidance.

Long-Term Sustainability and Industry Impact

Phase 3 outlines a long-term roadmap for autonomous debris-removal systems that integrate the AI decision-making models from Phase 2 into robotic platforms capable of safe, continuous operation in orbit. These systems represent a shift from reactive avoidance to proactive orbital stewardship.

Potential long-term impacts include:

- reducing the overall growth rate of debris populations,
- mitigating the compounded risk of collision cascades,
- extending the operational lifespan of satellites and constellations,
- lowering insurance premiums and operational costs for commercial operators,
- enabling safer human spaceflight and governmental missions.

By embedding decision-theoretic reasoning into robotic capture, tugging, or deorbiting systems, Phase 3 offers a technically viable path toward sustained, automated orbital hygiene.

This trajectory aligns closely with emerging ethical and governance frameworks such as the Astra Carta, the UNOOSA LTS Guidelines, and global sustainability initiatives. When combined with the cultural and policy foundations established in Phase 1, Phase 3 positions the space industry for a future in which orbital stewardship is operationalized and expected.

6.2 Challenges and Limitations

Despite its potential, the framework's implementation faces several challenges:

- Data and Tracking Limitations: High-uncertainty tracking data and untracked fragments complicate AI decision-making, necessitating improved global sensor networks.
- Algorithmic Complexity and Verification: Deploying AI in safety-critical orbital environments requires rigorous validation to ensure POMDPs and learning policies are provably safe.
- Policy and International Coordination: Non-binding guidelines, fragmented national regulations, and unclear liability for debris removal currently limit the operational viability of autonomous systems.

- Resource Constraints: The high cost of active debris removal missions and limited commercial incentives demand supportive policy and international collaboration to achieve broad adoption.

6.3 Contributions of This Work

This project makes three key contributions to the field of space sustainability:

1. Conceptual Innovation: It introduces a unified framework that bridges the typically separate domains of public awareness, AI-based reasoning, and long-term robotic mitigation.
2. Technical Insight: By adapting decision-making logic from domestic robotics to orbital debris, it highlights the universality of uncertainty-driven action planning and proposes a novel, intuitive pathway for AI-supported space operations.
3. Strategic Vision: It offers a coherent roadmap demonstrating how educational tools, public consciousness, policy evolution, and autonomous technology can be mutually reinforcing. This work lays the conceptual groundwork for future initiatives, including my long-term research vision of meaningful orbital stewardship.

In conclusion, the impact of this framework lies not only in its address of immediate technical challenges but in its cohesive, long-term vision for a responsible and sustainable space future.

7. Conclusion

This project proposes a holistic framework for addressing the increasingly urgent challenge of orbital debris by integrating three interconnected dimensions: public awareness, AI-driven decision-making, and long-term robotic mitigation. Rather than treating space debris as a purely technical or policy issue, the framework emphasizes that sustainable space stewardship requires a cultural foundation, one that makes the orbital environment visible, emotionally resonant, and widely understood. Without such awareness, the public cannot support ambitious policy measures, governments cannot justify large-scale mitigation investments, and industry cannot shift toward ethical, sustainable practices.

Phase 1 introduces *space debris consciousness* through gamified learning and interactive visualization tools. By enabling users to experience debris dynamics intuitively, this phase transforms an invisible threat into a relatable challenge. It mirrors the early strategies of environmental movements that shifted scientific facts into lived understanding and civic responsibility. In doing so, it establishes the cultural and educational footing upon which all subsequent phases depend.

Phase 2 builds on this foundation by adapting AI decision-making frameworks-rooted in robotics and uncertainty modeling-to orbital environments. By framing debris interactions as a Partially Observable Markov Decision Process, the project demonstrates how techniques originally developed for household tidying robots can optimize debris avoidance, reduce operator workload, and lay the groundwork for autonomous on-orbit decision systems. This technical contribution offers a pathway for integrating artificial intelligence into future space traffic management and sustainability operations.

Phase 3 extends the vision to its logical conclusion: autonomous robotic debris removal. As the orbital environment becomes more congested, active mitigation will be essential for maintaining long-term accessibility. By outlining how the decision-making frameworks of Phase 2 could scale into robotic action-including capture, maneuvering, and multi-object cleanup campaigns-this phase highlights a practical roadmap for transitioning from avoidance to remediation. This conceptual framework establishes the founding principles for this research agenda, whose initial technical development will focus on high-fidelity debris interaction simulators to train and validate the POMDP models described in Phase 2

Together, these phases form a coherent progression from understanding to intelligent action. They illustrate how cultural momentum, AI-driven reasoning, and robotics can reinforce one another to create a safer, more sustainable space environment. Although this project represents an early conceptual step, it lays the structural foundation for future work in policy, simulation, robotics, and public engagement. It also reflects a broader research vision: robots that tidy our environments, whether in homes or in orbit, guided by both human values and advanced autonomy.

In its entirety, the Space Debris Consciousness framework underscores a simple but profound insight: sustainable space futures begin with awareness, are strengthened by intelligent systems, and are realized through responsible action. This integrated approach offers a path toward preserving Earth's orbital commons for generations to come.

8. References

(To be developed with specific citations from)

- NASA Orbital Debris Program Office (ODPO)
- European Space Agency (ESA) Clean Space Initiative
- The Astra Carta Framework
- The Earthshot Prize
- POMDP & Reinforcement Learning Literature
- Relevant AA228 Course Materials

9. About the Researcher

Iyesomi Eigbe is a first-year graduate student in the Department of Aeronautics & Astronautics at Stanford University. Academically advised by Dr. Sigrid Elschot, her research focuses on orbital debris mitigation, AI-enabled decision-making, and policy for long-term space sustainability. This work, conducted for AA290: Research in Aeronautics and Astronautics, lays the conceptual groundwork for her future work which aims to develop robotic systems for sustaining environments on Earth and in orbit.